Cyber Security Internship Task-2

Internship Program: Cyber Security Internship by Future Interns

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Task-2: Phishing Email Detection & Awareness Report

Task Description

This task focuses on analyzing phishing emails and identifying malicious indicators using safe and non-intrusive techniques. The goal is to understand how phishing attacks operate and to create awareness guidelines that help users detect and avoid such threats.

Objective

The objective of this task is to:

- Analyze phishing email samples
- Identify phishing indicators
- Classify email risks
- Use basic security tools for validation
- Provide awareness and prevention guidelines
- Present findings in a structured report

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1. Introduction

Phishing is a social engineering attack where attackers impersonate trusted organizations to trick users into revealing sensitive information such as passwords, banking details, or OTPs.

These attacks are highly effective because they exploit human trust rather than technical vulnerabilities.

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2. Tools Used

The following tools were used during analysis:

- Google Admin Toolbox Messageheader
- MXToolbox
- Web Browser (for link inspection)
- MS Word / PDF (documentation)

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3. Email Samples

Email Sample 1 – Bank Phishing Email

From: authenticationmail@trust.ameribank7.com
To: johnsmith@email.com
Subject: **A new login to your bank account**

Bank of America

Dear account holder,

There has been a recent login to your bank account from a new divice:

IP address: 192.168.0.1

Location: Miami, Florida

4 new transactions have been made with this account since your last login.

If this was not you, please reset your password immediately with this link:

https://trust.ameribank7.com/reset-password

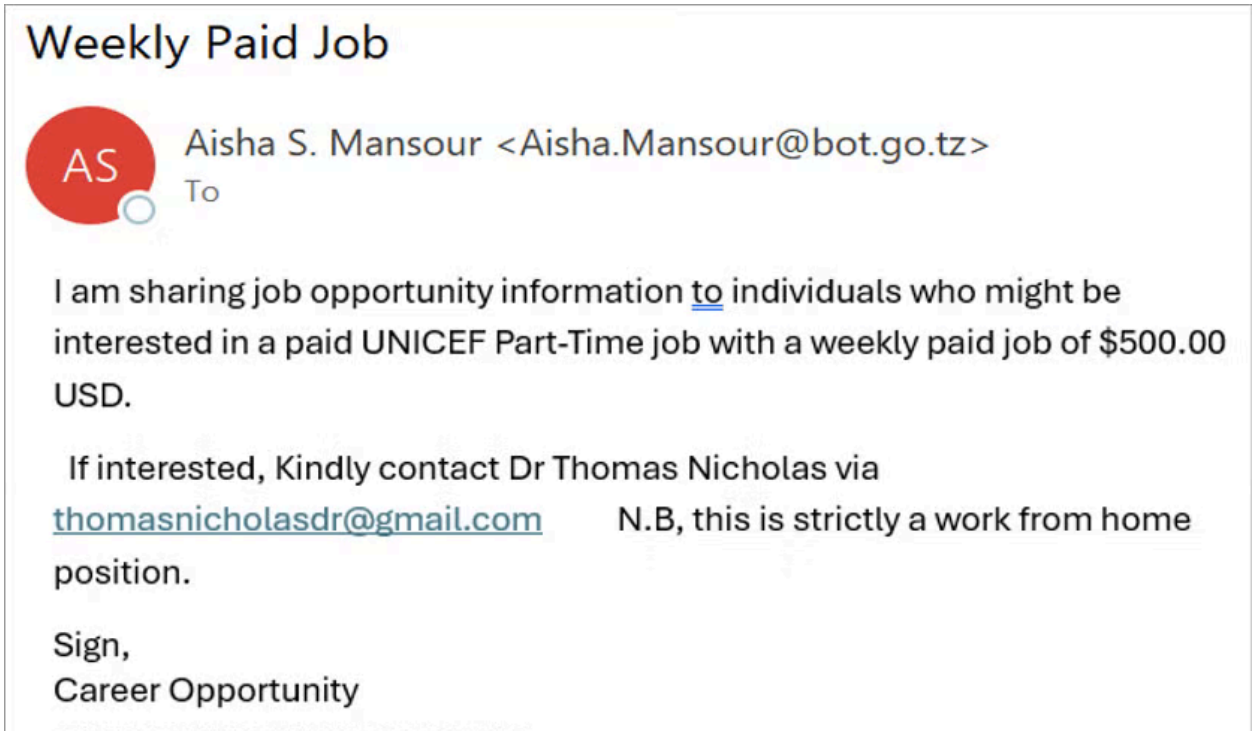
Thank you,

Bank America

From: authenticationmail@trust.ameribank7.com
Subject: A new login to your bank account

Content:
There has been a recent login to your account.
Reset your password immediately:
https://trust.ameribank7.com/reset-password

Email Sample 2 – Job Scam Email



From: Aisha.Mansour@bot.go.tz
Subject: Weekly Paid Job

Content Summary:
Offers a fake UNICEF job with $500 weekly payment and asks users to contact via Gmail.

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4. Methodology

The following steps were performed:

- Collected phishing email samples
- Inspected sender email addresses
- Analyzed suspicious links and domains
- Identified phishing indicators
- Performed basic domain checks using tools
- Documented findings with evidence

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5. Email Analysis

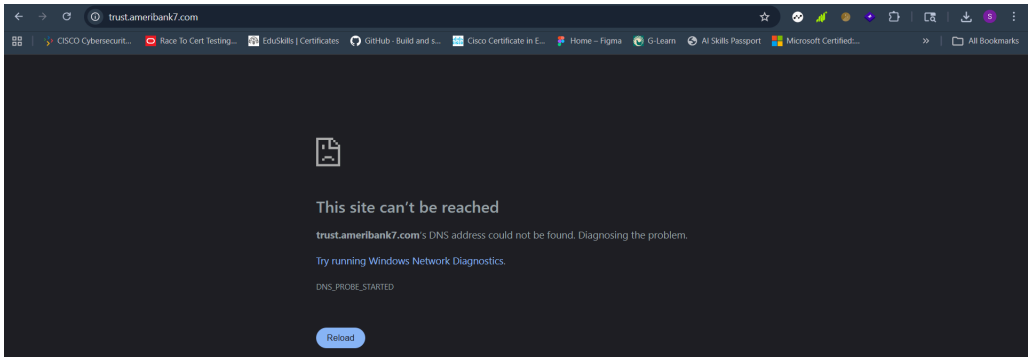
Email 1 Analysis (Bank Phishing)

Phishing Indicators

- Fake domain not associated with Bank of America
- Urgent instruction to reset password
- Suspicious link domain
- Generic greeting
- Grammar mistake (“divice”)
- Fake IP address (192.168.0.1)

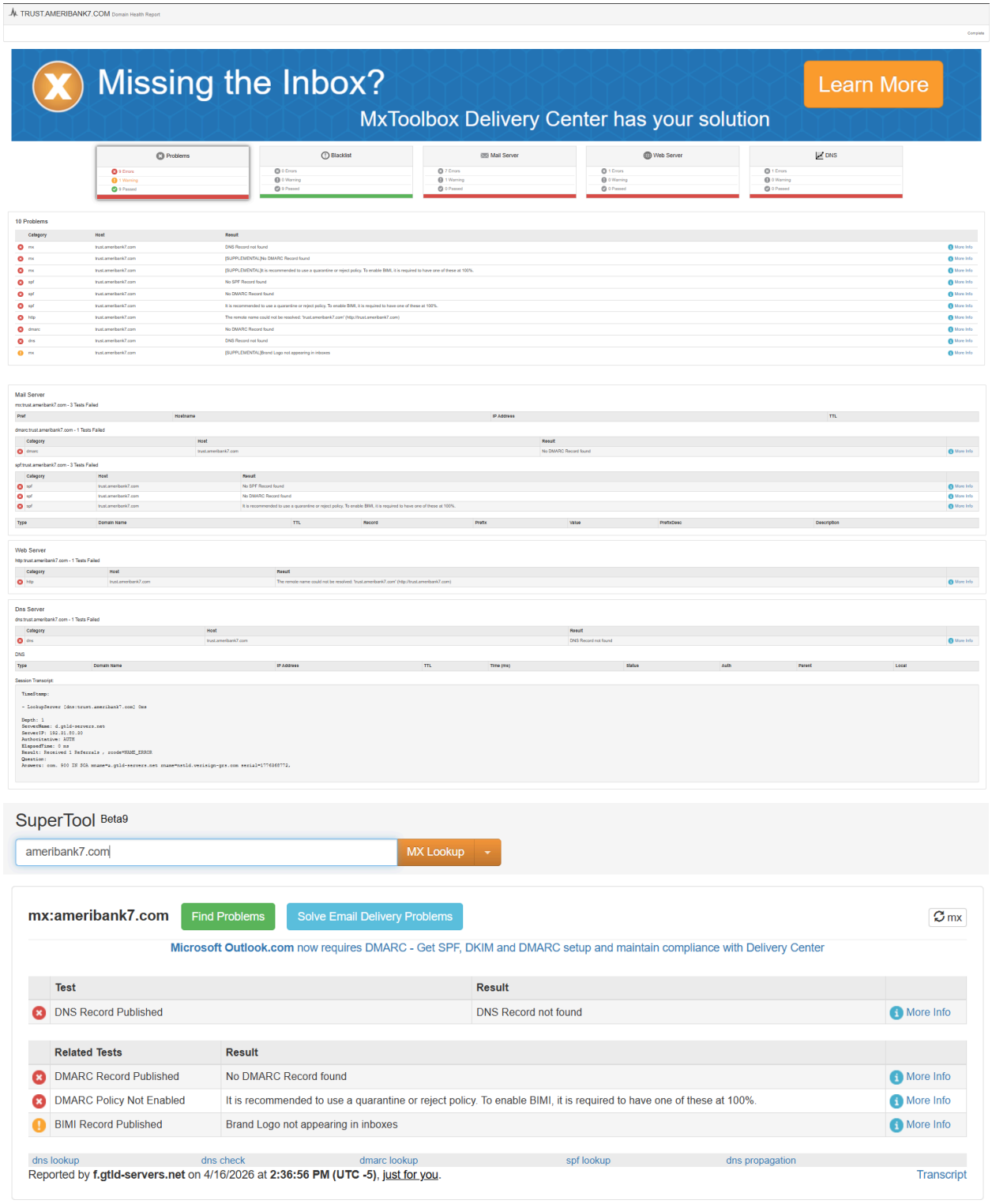
Link Analysis

The URL trust.ameribank7.com is not an official banking domain and is likely created to mimic a legitimate site.



Tool Usage (Domain Check)

The domain was analyzed using MXToolbox to verify legitimacy.



Risk Level - Phishing (High Risk)

Attack Explanation

The attacker impersonates a bank and creates urgency to trick users into clicking a fake link. Once credentials are entered, the attacker gains unauthorized access.

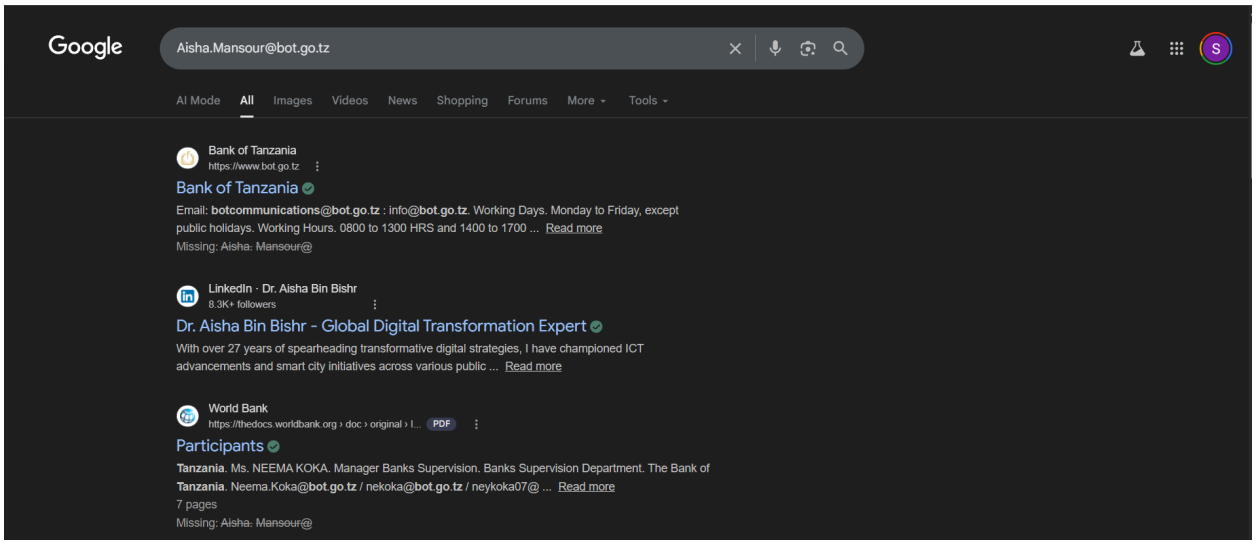
Email 2 Analysis (Job Scam)

Phishing Indicators

- Fake association with UNICEF
- Uses Gmail instead of official domain
- Unrealistic job offer ($500/week)
- Generic message
- External contact request

Link Analysis

The email redirects communication to a Gmail address instead of an official organizational domain.



Tool Usage (Domain Check)

Basic inspection was performed and domain credibility checked using MXToolbox

SuperTool Beta9

bot.go.tz | MX Lookup

mx:bot.go.tz

Find Problems | Solve Email Delivery Problems | mx

Microsoft Outlook.com now requires DMARC - Get SPF, DKIM and DMARC setup and maintain compliance with Delivery Center

Pref	Hostname	IP Address	TTL		
10	cluster2.eu.messagelabs.com	195.245.230.196 Google LLC (AS396982)	72 hrs	Blacklist Check	SMTP Test
20	cluster2a.eu.messagelabs.com	34.91.159.228 Google LLC (AS396982)	72 hrs	Blacklist Check	SMTP Test

	Test	Result
✔	DNS Record Published	DNS Record found

	Related Tests	Result	
✖	DMARC Record Published	No DMARC Record found	More Info
✖	DMARC Policy Not Enabled	It is recommended to use a quarantine or reject policy. To enable BIMI, it is required to have one of these at 100%.	More Info
!	BIMI Record Published	Brand Logo not appearing in inboxes	More Info

```
Your email service provider is "Symantec Connect" Need Bulk Email Provider Data?
```

dns lookup | dns check | dmarc lookup | spf lookup | dns propagation

Reported by **b.ns.tznic.or.tz** on 4/17/2026 at **6:02:32 AM (UTC -5)**, just for you. Transcript

smtp:bot.go.tz

Monitor This | Solve Email Delivery Problems | smtp

ARE YOU CONFIDENT that your email is getting through? FIND OUT WITH DELIVERY CENTER

Unable to connect after 15 seconds.

	Test	Result	
✖	SMTP Connect	Failed To Connect	More Info

Session Transcript:

```
Connecting to 196.46.101.39
4/17/2026 6:06:55 AM Connection attempt #1 - Unable to connect after 15 seconds.
[15.03 sec]


- LookupServer [smtp:bot.go.tz] 15255ms
```

Problems	Blacklist	Mail Server	Web Server	DNS
3 Errors	0 Errors	3 Errors	0 Errors	0 Errors
9 Warning	0 Warning	6 Warning	0 Warning	3 Warning
423 Passed	352 Passed	50 Passed	8 Passed	13 Passed

12 Problems

	Category	Host	Result	
✖	dmarc	gmail.com	DMARC Quarantine/Reject policy not enabled	More Info
✖	spf	gmail.com	It is recommended to use a quarantine or reject policy. To enable BIMI, it is required to have one of these at 100%.	More Info
✖	mx	gmail.com	[SUPPLEMENTAL]It is recommended to use a quarantine or reject policy. To enable BIMI, it is required to have one of these at 100%.	More Info
!	mx	gmail.com	[SUPPLEMENTAL]Brand Logo not appearing in inboxes	More Info
!	dns	gmail.com	SOA Serial Number Format is Invalid	More Info
!	dns	gmail.com	SOA Refresh Value is outside of the recommended range	More Info
!	dns	gmail.com	SOA Expire Value out of recommended range	More Info
!	smtp	gmail-smtp-in.l.google.com	Reverse DNS does not match SMTP Banner	More Info
!	smtp	alt3.gmail-smtp-in.l.google.com	Reverse DNS does not match SMTP Banner	More Info
!	smtp	alt4.gmail-smtp-in.l.google.com	Reverse DNS does not match SMTP Banner	More Info

Risk Level - Phishing / Scam (High Risk)

Attack Explanation

The attacker uses a fake job offer to lure victims into sharing personal details or making payments.

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6. Evidence Collection

The following evidence was collected:

- Email screenshots (phishing samples)
- Suspicious link inspection screenshots
- Domain analysis results using MXToolbox
- Header analysis (if available) using Google Admin Toolbox Messageheader.

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7. Findings

Indicator	Email 1	Email 2
Fake Domain	✅	✅
Urgency	✅	❌
Suspicious Link	✅	❌
Generic Content	✅	✅
Grammar Issues	✅	❌
Scam Offer	❌	✅
Risk Level	🔴 High	🔴 High

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8. Impact Analysis

These phishing attacks can lead to:

- Account compromise
- Financial fraud
- Identity theft
- Unauthorized access
- Loss of sensitive information

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9. Risk Classification

Severity	Description
High	Confirmed phishing attack
Medium	Suspicious email
Low	Legitimate email

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10. Prevention Guidelines

✔ Do’s

- Verify sender email address
- Check official domains
- Hover over links before clicking
- Report suspicious emails

❌ Don’ts

- Don’t click unknown links
- Don’t share passwords or OTPs
- Don’t trust urgent messages blindly
- Don’t respond to fake job offers

11. Conclusion

This report demonstrates how phishing attacks exploit trust using fake domains, urgency, and attractive offers. User awareness and careful verification are critical in preventing such attacks.

This task helped develop practical skills in phishing detection and cybersecurity reporting.

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12. References

- OWASP Phishing Guidelines
- MXToolbox
- Google Admin Toolbox Messageheader